Aryaman Maithani | Curriculum Vitae

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Education

University of Utah — **Salt Lake City, USA**
Ph.D. student in Mathematics — *2022–present*
Advisor: Anurag K. Singh

Indian Institute of Technology Bombay — **Mumbai, India**
B.S. in Mathematics with Honors — *2018–2022*

Awards/Fellowships

T. Benny Rushing and Gail T. Rushing Fellowship — **University of Utah**
Research fellowship in recognition of academic excellence and research promise — *2026*

Simons Dissertation Fellowship in Mathematics — **Simons Foundation**
Research fellowship for outstanding doctoral students during the final stages — *2025–2027*

Pure Mathematics Fellowship — **University of Utah**
Department research support — *Fall 2024, Fall 2025, Spring 2027*

Institute Gold Medal — **IIT Bombay**
For outstanding academic performance among graduating students — *2022*

Institute Academic Prize — **IIT Bombay**
For outstanding academic performance among mathematics majors — *2020, 2021*

Research

Commutative algebra: combinatorial commutative algebra, homological algebra, invariant theory, local cohomology, rings of differential operators.

Refereed publications

9. *Abelian extensions of equicharacteristic regular rings need not be Cohen-Macaulay*, arXiv
with Anurag K. Singh and Prashanth Sridhar
Proceedings of the American Mathematical Society **154**:11, 4667–4672 (2026)
DOI: 10.1090/proc/17803

8. *Polynomial invariants of classical subgroups of* GL_2*: Conjugation over finite fields* arXiv
Journal of Pure and Applied Algebra **230**:8, article no. 108307 (2026)
DOI: 10.1016/j.jpaa.2026.108307

7. *Homological properties of invariant rings of permutation groups* arXiv
Proceedings of the American Mathematical Society **154**:8, 3257–3272 (2026)
DOI: 10.1090/proc/17664

6. *Splitting the difference: Computations of the Reynolds operator in classical invariant theory* arXiv
Bulletin of the London Mathematical Society **58**:1, article no. e70175 (2026)
DOI: 10.1112/blms.70175

5. *Minimal cellular resolutions of powers of graphs*, arXiv
with Trung Chau and Tài Huy Hà
Electronic Journal of Combinatorics **33**:1, article no. P1.25 (2026)
DOI: 10.37236/13606

4. *Edge ideals with linear quotients and without homological linear quotients*, arXiv
with Trung Chau and Kanoy Kumar Das
Mediterranean Journal of Mathematics **23**, article no. 31 (2026)
DOI: 10.1007/s00009-025-03028-5

3. *Monomial ideals with minimal generalized Barile–Macchia resolutions*, arXiv
with Trung Chau and Tài Huy Hà
Vietnam Journal of Mathematics **54**, 307–324 (2026)
DOI: 10.1007/s10013-025-00773-8

2. *The Scarf complex of squarefree powers, symbolic powers of edge ideals, and cover ideals of graphs*, arXiv
with Trung Chau and Nursel Erey
Communications in Algebra **54**:3, 982–1001 (2026)
DOI: 10.1080/00927872.2025.2544296

1. *Linear quotients of connected ideals of graphs*, arXiv
with H. Ananthnarayan and Omkar Javadekar
Journal of Algebraic Combinatorics **61**, article no. 34 (2025)
DOI: 10.1007/s10801-025-01395-6

Preprints

3. *On the b-invariant of a normal graded ring*,
with Anurag K. Singh and Kei-ichi Watanabe arXiv:2609.32003 (2026)

2. *The arithmetic rank of nullcones of classical invariant rings*,
with Manav Batavia and Kesavan Mohana Sundaram arXiv:2608.29445 (2026)

1. *Congruence modules and Wiles defects of determinantal rings of maximal minors*,
with Kashif Khan arXiv:2604.06446 (2026)

Talks

Seminars

Algebra Seminar **Syracuse University**
Homological properties of invariant rings of permutation groups *October 2026*

AMS 2026 Spring Central Sectional Meeting **North Dakota State University**
Polynomial invariants of GL_2: Conjugation over finite fields *April 2026*

Commutative Algebra Seminar **University of Nebraska-Lincoln**
Polynomial invariants of GL_2: Conjugation over finite fields *February 2026*

Commutative Algebra Seminar **Purdue University**
Polynomial invariants of GL_2: Conjugation over finite fields *September 2025*

Commutative Algebra Seminar **UC San Diego**
Polynomial invariants of GL_2: Conjugation over finite fields *February 2025*

Commutative Algebra Seminar **University of Utah**
Polynomial invariants of GL_2: Conjugation over finite fields *February 2025*

Commutative Algebra Seminar **IIT Bombay**
Invariant theory of commutative rings *August 2024*

Commutative Algebra Seminar **University of Utah**
Linear quotients of connected ideals of graphs *March 2024*

Dualities in Topology and Algebra **ICTS, India**
Gorenstein rings *May 2023*

Graduate Student Seminars

BIKES, Commutative Algebra Graduate Student Seminar **University of Utah**
On the Cohen–Macaulayness of abelian extensions of regular local rings *September 2026*

BIKES, Commutative Algebra Graduate Student Seminar **University of Utah**
On invariant rings of permutation groups *January 2026*

Ideal Conversations, Commutative Algebra Graduate Student Seminar **Purdue University**
Invariant theory of commutative rings *September 2025*

BIKES, Commutative Algebra Graduate Student Seminar — **University of Utah**
Splittings in Classical Invariant Theory — *August 2025*

BIKES, Commutative Algebra Graduate Student Seminar — **University of Utah**
Examples of badly-behaved rings — *August 2024*

BIKES, Commutative Algebra Graduate Student Seminar — **University of Utah**
Linear quotients of connected ideals of graphs — *February 2024*

Commutative Algebra Graduate Student Seminar — **University of Michigan**
Invariant theory of commutative rings — *October 2023*

BIKES, Commutative Algebra Graduate Student Seminar — **University of Utah**
Invariant theory of commutative rings — *September 2023*

General Math Talks

Graduate Student Colloquium — **University of Utah**
Shut your π hole! — *March 2024*

Math Circle — **University of Utah**
Fun with graphs — *March 2023*

Graduate Student Colloquium — **University of Utah**
Chaos theory — *February 2023*

Teaching Experience

Instructor of Record

MATH 2200: Discrete Mathematics
University of Utah — *Fall 2026*

MATH 1220: Calculus II
University of Utah — *Spring 2025, Summer 2024*

MATH 1090: Business Algebra
University of Utah — *Spring 2024*

MATH 1060: Trigonometry
University of Utah — *Fall 2023*

MATH 1010: Intermediate Algebra
University of Utah — *Summer 2023*

Graduate Teaching Assistant

MATH 2250: Differential Equations and Linear Algebra
University of Utah — *Spring 2023, Fall 2022*

Teaching Assistant for Workshops

Recent Developments in Commutative Algebra
IIT Dharwad — *24–29 June 2025*

Pre-REU 2025: Introducing undergraduates to research
University of Utah — *6–30 May 2025*

Grader for Graduate Courses

MATH 6520: Algebraic Topology
University of Utah — *Spring 2024*

MATH 6310: Modern Algebra 1
University of Utah — *Fall 2023*

MATH 6320: Modern Algebra 2
University of Utah — *Spring 2023*

Service/Outreach

- **University of Utah** **Salt Lake City, Utah, USA**
  Graduate Student Advisory Committee
  - 2024–2025: **GSAC Co-chair.**
    Liaison between graduate students and department.
  - 2023–2025: **Recruitment committee.**
    Welcome prospective graduate students and organize meetings with faculty for them.

Technical skills

Python, SageMath, Macaulay2, Magma